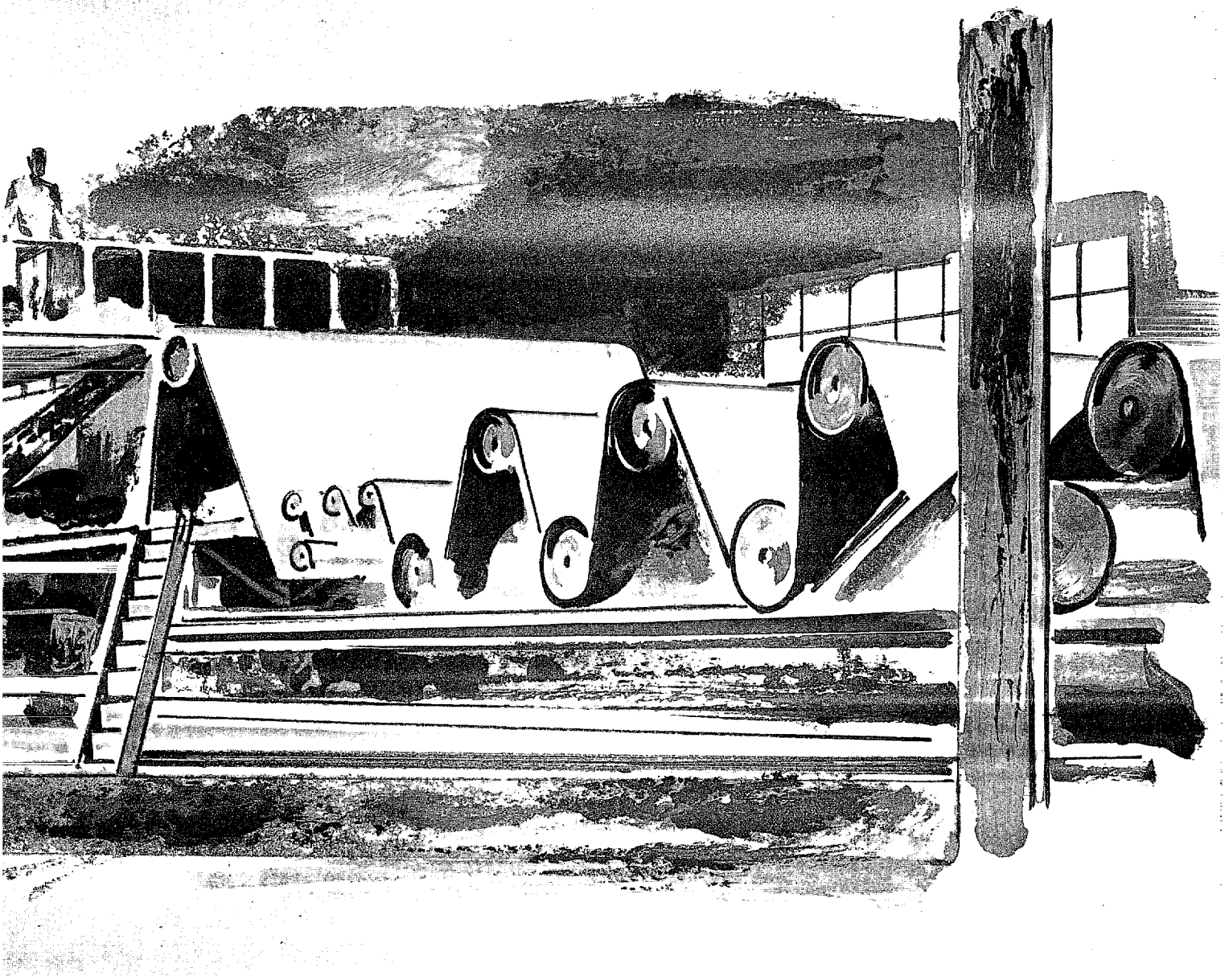
GEK-7875

INSTRUCTIONS

MECHANICAL BRAKE WINDER CONTROL

FOR DRIVES WITH SILCOMATIC TYPE POWER SUPPLY



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GENERAL ELECTRIC

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These instructions do not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation or maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to the General Electric Company.

MECHANICAL BRAKE WINDER CONTROL

INTRODUCTION

GENERAL

These instructions describe the operation, adjustments, and maintenance of the electrical control equipment for a surface-driven, mechanical brake winder used in paper mill applications. It is assumed that the reader is familiar with the mechanical equipment and general operation of paper mill winders.

The description, illustrations and circuits used in these instructions are for a typical control. The elementary diagram and other drawings furnished with a specific equipment will show and explain the optional or additional features furnished for that specific equipment.

Figure 1 shows, in simplified circuit form, how the various electrical control components and machines are connected together as a system. PS5 is the power supply module that supplies plus and minus 50 volts DC to the other regulator modules. RRA is the rate reference module that supplies the voltage (or speed) reference signal to the regulator. VSA is the main regulator module that compares the voltage (or speed) reference signal to the Silcomatic output voltage (or motor speed) and changes the control signal to the Silcomatic as required to control the Silcomatic output to the desired level. CIA is an isolation amplifier that isolates and amplifies the current signal received from the ammeter shunt. This signal is used for current limit purposes. REC is a transformer - rectifier combination that supplies 250V DC constant potential for the motor field, control relays, etc.

MAGNETIC CONTROL CIRCUITS

GENERAL

Figure 2 shows the magnetic control circuits for a typical mechanical brake winder. Note that the handwheel controller is the main controlling device since it is used to start, stop and control the speed of the drive.

DRIVE START

Before the drive can be started, all overcurrent, overload and other protective devices must be in their reset position. Also the drive motor must be stopped, as sensed by the motor speed relay CP being deenergized. The drive is started by turning the handwheel to the ON position. This energizes the run relay R, the main line contactor M1, and the contactor auxiliary relay MX.

SPEED CONTROL

Drive speed is adjusted by varying a potentiometer which is mechanically turned by the handwheel controller (see Figure 5 for location of the potentiometer in the circuit). A detailed discussion of speed control is included in the Regulator section of these instructions.

DRIVE STOP

1. Normal Stop: A normal stop is initiated by turning the handwheel controller to the off position. The motor will decelerate at the timed rate set by

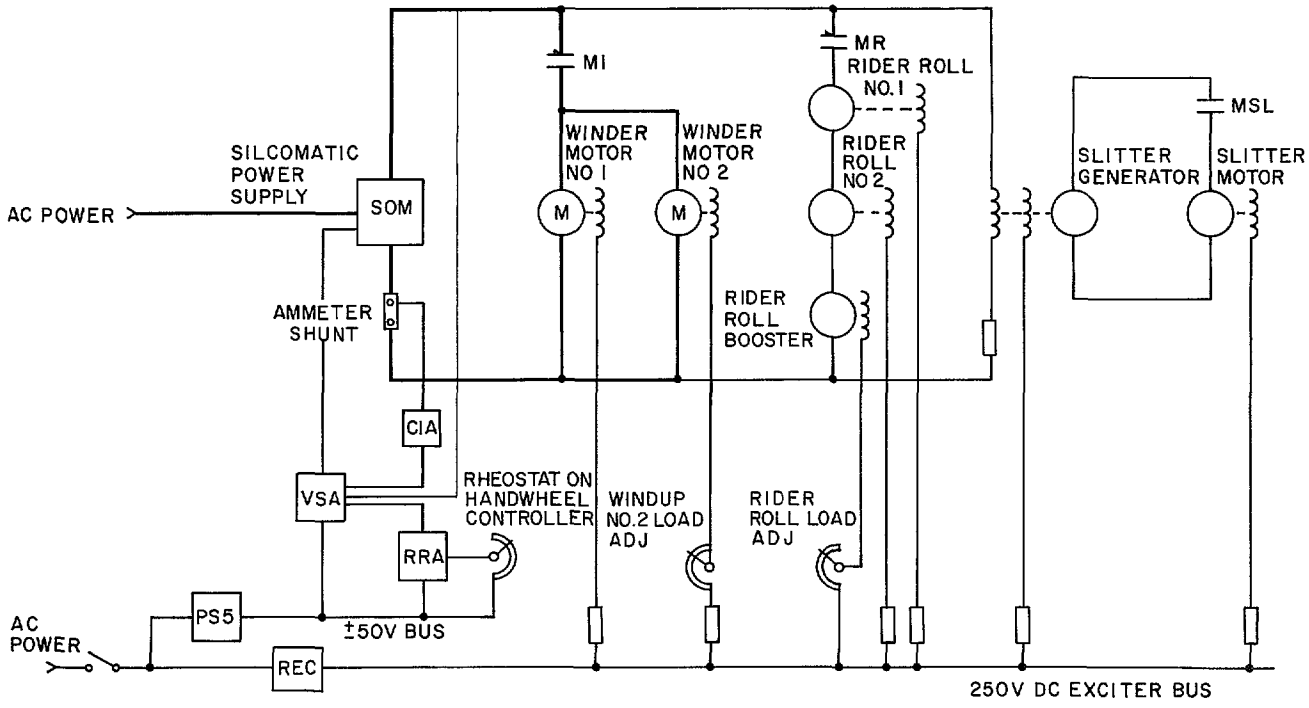
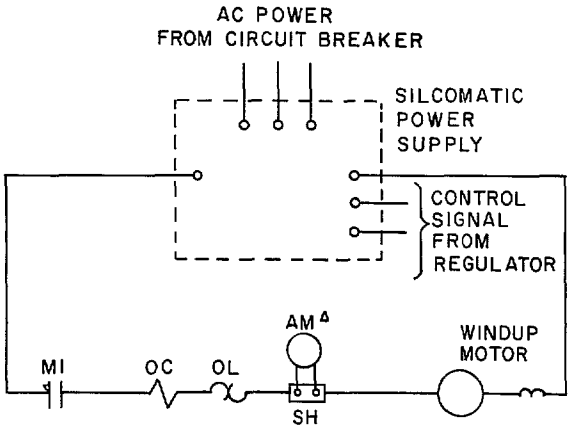


Fig. 1. Simplified circuit diagram of typical mechanical brake winder



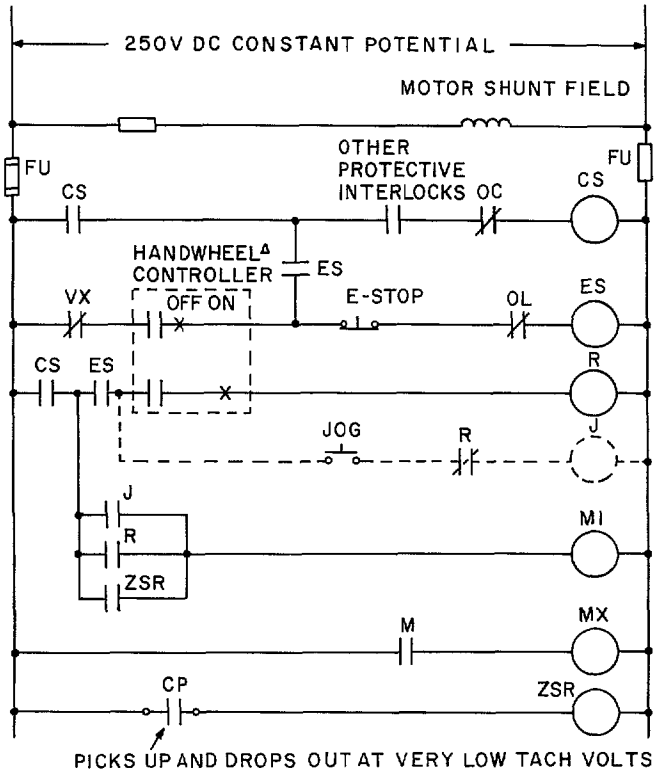


Fig. 2. Magnetic control circuit for typical mechanical brake winder

the regulator rate reference circuit (or under current limit control if the rate is set faster than the motor or Silcomatic rating will permit). When the motor speed has decreased to a very low value the speed relay CP will drop out, the zero speed relay ZSR will drop out, and the line contactor M1 will open.

2. Emergency Stop: An emergency stop is initiated by any action that deenergizes the emergency stop relay ES, such as the operation of a motor overload or pressing the emergency stop pushbutton. The stopping sequence is now similar to that of a normal stop except that the speed reference signal is immediately changed to zero so the drive always stops under current limit control. On some drives the current limit circuit may also be recalibrated to allow a higher than normal stopping current and therefore, a faster stop.

3. Coast Stop: A coast stop is initiated by any action that deenergizes the line contactor M1 or the coast stop relay CS. Motor overcurrents or operation of the protective interlocks in the coast stop coil circuit are typical of such an action. Under coast stop conditions there is no motor armature current and hence no braking action.

PUSHBUTTON OPERATION (OPTIONAL)

When pushbutton control is specified, the functions of the handwheel controller are replaced by start and stop pushbuttons and a separate, manually operated, rheostat.

JOG OPERATION (OPTIONAL)

The circuit for this function is shown in broken lines in Figure 2. The function utilizes a pushbutton and relay to run the drive at a low (0 to 5 percent) speed. The jog speed level is adjusted by a rheostat in the speed regulator circuit. The drive will run only while the jog pushbutton is held depressed.

RIDER ROLL CONTROL (OPTIONAL)

The circuit for this function is shown in Figure 3. If the selector switch is in the ON position, the rider roll, through relay ZSR, will start and stop with the main drive. The rider roll drive will not start unless the winder motor speed is below the pickup value of the speed relay CP and its auxiliary relay ZSR. The rider roll drive can be stopped independently of the main winder drive by turning the selector switch to the OFF position.

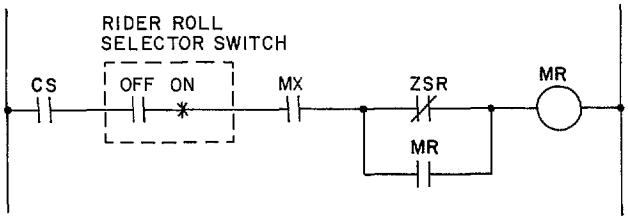


Fig. 3. Control circuit for rider roll

SLITTER CONTROL (OPTIONAL)

The circuit for this function is shown in Figure 4. Operation of this circuit is similar to that for the rider roll with the exception that the main winder drive does not have to run in order to have the slitter drive run.

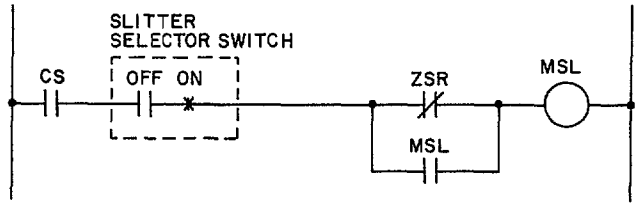


Fig. 4. Control circuit for slitter

REGULATOR

GENERAL

Figure 5 shows a simplified diagram of the basic regulator. Most of the elements of the regulator circuits are physically contained in the following modules:

PS5	Power Supply Module	GEK-7906
RRA	Rate Reference Module	GEK-7905
VSA	Voltage and Speed Amplifier Module	GEK-7907

The description and operation of these modules is explained in the instruction books identified at the right in the above list. However, the following brief explanation of the overall regulator action will help to tie the various pieces together.

There are five significant modes of operation for the system:

1. Standstill (when line contactor is open)
2. Normal Operation
3. Current Limit Operation
4. Emergency Stop Operation
5. Jog Operation

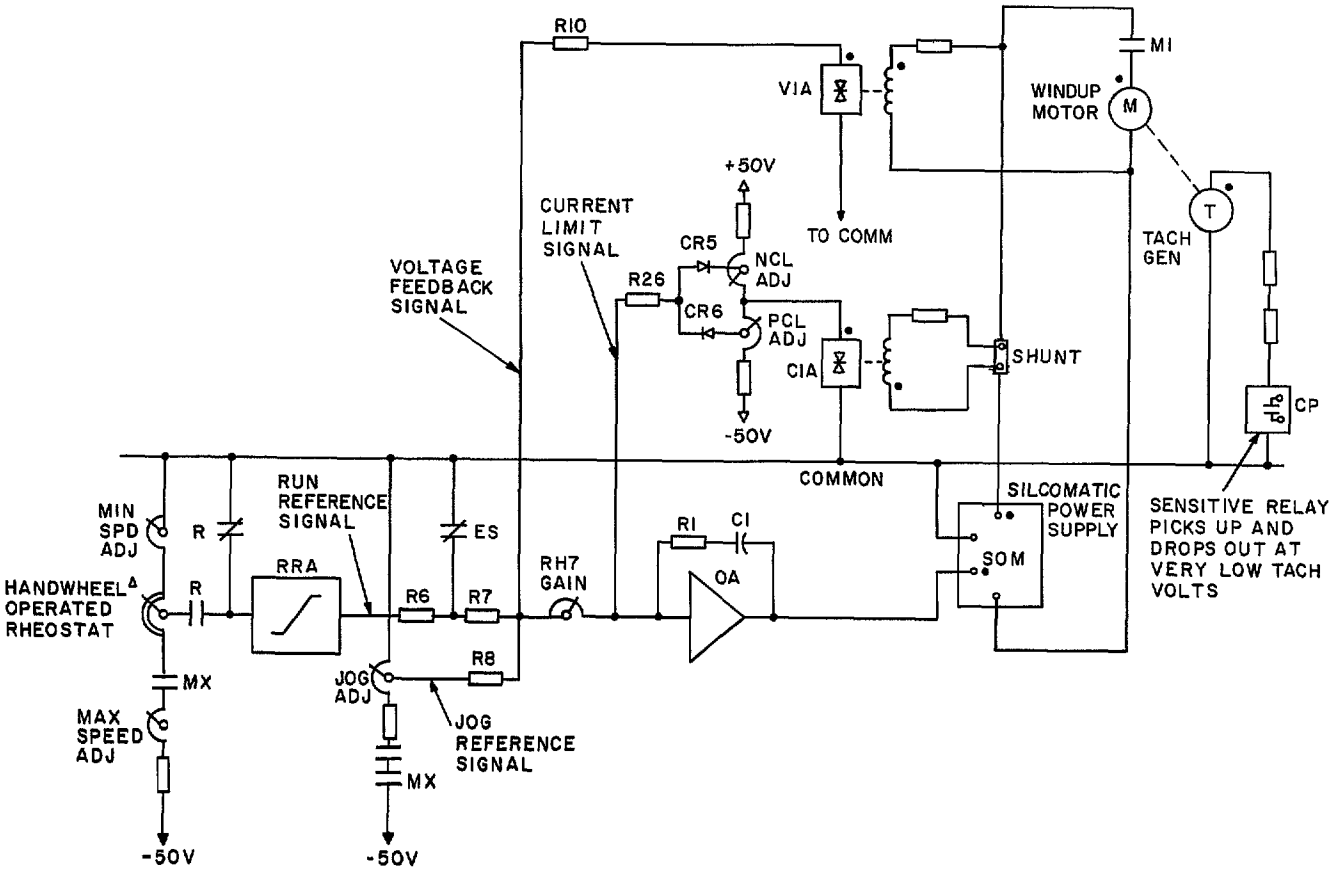


Fig. 5. Simplified circuit of basic regulator

STANDSTILL

When the drive is at standstill the run relay R, the jog relay J, the line contactor M1 and the contactor auxiliary relay MX are dropped out. In this mode the run reference signal and the jog reference signal are zero, so a zero reference signal is applied (through Gain rheostat RH7) to the operational amplifier OA. The Silcomatic output voltage is fed back to the operational amplifier via the voltage isolation amplifier VIA and resistor R10. The polarities of the various elements are such that any voltage at the Silcomatic output causes the operational amplifier to apply a voltage to the Silcomatic input. This causes the Silcomatic voltage to decrease. The operational amplifier gain is extremely high (approximately 0.006 microamps input current produces full output of 25 volts) so the Silcomatic voltage is held to practically zero.

NORMAL OPERATION

When the drive is energized in the normal (run) mode, the run relay R, the line contactor M1 and the contactor auxiliary relay MX are energized. A portion of the voltage across the handwheel operated rheostat is applied to the input of the Rate Reference Amplifier module RRA. This module contains a transistorized ramp generator whose output voltage gradually increases (or decreases) to the level of the input voltage. (See the RRA instruction book GEK-7905 for more complete information on this module.) The output of the rate reference amplifier RRA is applied to the operational amplifier OA input via resistors R6 and R7 as a run reference signal. This reference signal causes the operational amplifier output voltage and the Silcomatic voltage to increase and the motor to rotate. The Silcomatic voltage is fed back to the operational amplifier via the voltage isolation amplifier VIA and resistor R10 (similar to the Standstill operation). Its polarity is then at a level that tends to reduce the operational amplifier and Silcomatic output voltages. When the feedback signal strength matches the reference signal strength the regulator is balanced and the Silcomatic voltage remains steady.

If the operator moves the handwheel operated rheostat to a higher setting, the input to the rate reference amplifier RRA is increased. When this occurs the RRA output and reference signal gradually increase and the operational amplifier output and Silcomatic output increase until the feedback signal again matches the reference signal. Thus the Silcomatic voltage is regulated to a value proportional to the setting of the operators handwheel operated rheostat.

When the handwheel is turned to its off position, the run relay R drops out and zero volts are applied to the input of the rate reference amplifier RRA. The RRA output gradually goes to zero, and the Silcomatic voltage follows the reference signal down. The windup motor RPM decreases with the Silcomatic voltage. When the motor speed (and therefore tach voltage) gets to a very low value, the speed relay CP drops out and deenergizes the line contactor M1 and the drive is returned to the Standstill mode.

CURRENT LIMIT OPERATION

The output current of the Silcomatic is sensed by the ammeter shunt. The voltage across the shunt is amplified by the current isolation amplifier CIA and when the output voltage of CIA exceeds the voltage set on the positive current limit adjust pot PCL (or the negative current limit adjust pot NCL). A signal is then fed through the diode CR6 (or CR5) and resistor R26 to the operational amplifier in a direction to prevent the Silcomatic voltage from going any higher (or lower), thus limiting the Silcomatic current to a specific value. For example, assume the Silcomatic output current is positive (current leaves Silcomatic terminal that has dot beside it in Figure 5). This causes CIA output to be positive with respect to common. However, as long as this voltage is less than the negative voltage added to the circuit by the positive current limit pot PCL, current flow in the circuit will be blocked by diode CR6. When the Silcomatic output current is high enough to raise the PCL slider slightly positive with respect to common, a current will flow through diode CR6 and resistor R26 to the operational amplifier input. This signal effectively adds to the voltage feedback signal and helps balance the reference signal. The resistors in the circuits are chosen so that the current limit signal is very strong. It is thus capable of completely controlling the Silcomatic output once the Silcomatic output current causes CR6 (or CR5) to conduct. The point at which this occurs is adjustable by adjusting the pots PCL and NCL.

EMERGENCY STOP OPERATION

When an emergency stop is initiated, the emergency stop relay ES and the run relay R drop out. A normally closed ES interlock immediately reduces the run reference signal to zero by connecting the reference circuit between resistors R6 and R7 to zero (common). The voltage feedback signal tries to reduce the Silcomatic voltage to zero, but the current limit circuit will restrain the rate of Silcomatic voltage decrease to a safe value. When the motor speed is very low the speed relay CP drops out (as in a normal stop), the line contactor is opened, and the drive returns to the Standstill mode.

JOG OPERATION (OPTIONAL)

When a jog is initiated (by pressing the JOG pushbutton) the jog relay J, the line contactor M1 and the contactor auxiliary relay MX pick up. A jog reference signal is applied via resistor R7 to the operational amplifier. The operation is now the same as for normal operation except the jog reference signal is used instead of the normal run reference signal.

SPEED REGULATOR (OPTIONAL)

In some applications an optional speed regulator may be supplied. The speed regulator uses the same regulator modules and the same basic circuits as the regulator previously described. The major difference is that whenever the contactor auxiliary relay is energized one of its contacts removes the voltage feedback signal and applies, in its place, a speed feedback signal which is obtained from the tachometer generator. A more complete description of this circuit, plus the stabilizing circuits used for the speed regulator, is given in the VSA module instruction book GEK-7907.

INSTALLATION AND TUNE-UP

PRELIMINARY INSPECTION

1. Visually inspect the equipment for shipment damages.

2. If the equipment is to be stored before being used, provide adequate protection against the elements and other contaminating conditions.

3. If the equipment has been stored or unused for prolonged periods, be sure all equipment is dry and clean before energizing.

4. Manually move all mechanical devices and check for binding or misalignment.

5. Check to certify that all interconnecting wires and remote devices agree with the elementary diagram.

6. Check shielded wires to confirm that the shields have been tied down in accordance with the interconnection drawing.

7. Check terminals for loose connections, electrical clearance, etc.

8. Make sure all contacts are clean and properly positioned.

9. Check motor armatures and fields for proper ohms and grounds. In checking for grounds avoid subjecting the control equipment to voltages which might cause damage to the semiconductor components.

10. Make certain that the voltage and current feedback signals are being extracted from the proper points on the Silcomatic and the ammeter shunt.

11. Be sure that adequate electrical, thermal, and mechanical clearances have been maintained.

12. Establish effective communication between control panels and operator's desk prior to actually tuning-up the equipment.

TUNE-UP PROCEDURE

General

Experience has indicated that following certain procedures will result in a more trouble-free and less expensive installation. Some of these procedures are listed below. The following instructions should not be interpreted as a representation of all conditions which might be encountered or need to be considered during any start-up. The actual start-up procedure should be established by those responsible, giving due regard to safety of personnel and protection of the electrical and mechanical equipment involved. The following instructions assume that the rotating machines and other remote mounted equipment are in proper working order. It is also assumed that the preliminary inspection has been completed and the system is free from grounds, haywires, etc.

Initial Conditions

Before applying power to the control equipment, open the control power disconnect switches and disconnect one motor armature circuit cable to each motor at the control panel. Also, disconnect one field lead to each motor and booster at the control panel.

Exciter Checkout

Energize the 250 volt DC Constant Potential Exciter and see that it produces 250 ± 10 volts DC.

Magnetic Circuits

Close the control power switch and check that the pushbutton, relay and contactor coil circuits are

functioning properly. Use the elementary diagram to insure that all circuits are checked.

Motor and Booster Field Currents

a. Reconnect the motor fields and see that the motor field currents are approximately correct and that any special field circuits such as field economy, field rheostats, etc. are operating properly. Refer to the machine data sheet of the elementary diagram for approximate field current values.

b. If booster generators are being used, reconnect their fields and see that the field resistors are set to produce the desired booster output voltage.

c. The above adjustments should be approximate only. Final adjustments will be made later.

Power Supply Module

Set up the power supply module as described in the Installation Adjustment section of the PS5 module instruction book GEK-7906. If the windup tachometer generator voltage is 250 volts or less connect the tachometer lead to terminal A1M-11. If the voltage is more than 250 volts connect the tachometer lead to terminal A1M-9.

Rate Reference Module

Set up the Rate Reference module as described in the Installation Adjustment section of the RRA module instruction book GEK-7905. Operate the run relay and contactor auxiliary relay interlocks by hand if necessary to permit checking the accel and decel times, etc. Temporarily set the accel and decel times for 100 seconds.

Voltage Regulator

Set up the VSA module in its "test" mode. That is, move the VSA test card from slot I to slot J. Test and set up the VSA as described in the Trouble Shooting section of the VSA instruction book GEK-7907. Note that the operation of the VSA module in its "test" mode permits manual control (by means of the test potentiometer RH4 in the VSA module) of the VSA output voltage. This feature can be used to check out the operation of the Silcomatic Power Supply. Return the VSA test card back to its normal position.

Maximum and Minimum Speed

Reconnect the motor armature leads, set the current limit pots PCL and NCL on the VSA module to their mid position, and start the drive at a slow speed. Check to see that the motor rotation and the tachometer generator polarity are correct (positive relative to common). Gradually accelerate the drive until the handwheel operated rheostat is in its maximum speed position. Check the Silcomatic voltage and if necessary adjust the Max Ref pot on the RRA module to obtain rated Silcomatic voltage. Decelerate the drive to the minimum speed position on the handwheel operated rheostat and adjust the Min Ref pot in the RRA module to obtain the minimum speed desired by the operators.

Speed Relay

Stop the drive and see that the speed relay CP drops out at less than 1 percent of rated drive speed.

Current Limit

While the drive is operating, observe the Silcomatic output current and adjust the gain of the current isolation amplifier CIA (by means of gain pot built into CIA) to obtain 7 1/2 to 8 volts output of CIA for rated Silcomatic current. Decrease the accel and decel times to a very low value and adjust the current limits pots PCL and NCL on the VSA module to obtain the desired current limit values. This is normally less than 200 percent of Silcomatic rated current.

Jog Speed

Initiate a jog operation and adjust the Jog Ref adjust potentiometer in the VSA module to obtain the desired jog speed.

Final Motor and Booster Field Adjustments

Make final adjustments to the windup motor field resistors so that when the Silcomatic is at rated output voltage the windup roll surface speed is equal to machine rated sheet speed. If the #2 Windup Drum has a load adjust rheostat, this rheostat should be approximately 2/3 shorted out during this adjustment. Make final adjustments to the rider roll motor field resistors so that when the Silcomatic is at rated voltage (and the rider roll booster generator is at minimum voltage) the rider roll surface speed is equal to machine rated speed. If two rider roll motors are used in a series connection the field resistors should be set to assure approximately equal voltage across each motor.

Make final adjustments to the rider roll booster field resistors so that the booster causes rated motor current to flow (when the rider roll is against the paper on the windup roll) when the operators load adjust rheostat is in its maximum load position.

Make final adjustments to the slitter generator field resistors to obtain the desired minimum voltage at zero Silcomatic volts and the desired maximum volts at rated Silcomatic volts. Make final adjustments to obtain the desired slitter speed at maximum voltage of the slitter generator.

All of the above field resistor adjustments should be made when the motors and boosters are near their operating temperature. If the adjustments are made when the machines are cool, remember that the field resistance will increase (as much as 25 percent) as the field temperature increases from room temperature to machine rated temperature.

Speed Regulator

If a speed regulator is furnished the basic instructions outlined above apply, except as follows:

a. Set up the VSA module per the instruction in the VSA instruction book.

b. Adjust the windup motor field resistors so that the Silcomatic voltage is slightly below rated voltage when the windup is operating fully loaded at rated sheet speed.

c. Adjust the rider roll, slitter, etc. fields after step b is completed.

d. Adjust the speed of response (stability) of the speed regulator (per VSA instruction book) to obtain a moderate speed of response. Extremely fast response is neither necessary nor desirable.

Final Adjustment of Accel and Decel Times

Set the acceleration and deceleration times by adjusting the accel and decel pots in the RRA module so that the purchaser's tension system can keep up with the change in sheet speed when the windup has a small roll and the unwind has a full roll.

MAINTENANCE

GENERAL

Periodic inspections and maintenance are important to continuous operation. Regular inspections should be performed so that replacements or repairs can be made before a failure occurs. Costly downtime can be reduced by following three basic rules:

1. Prepare a preventive-maintenance program consisting of items to be checked and recorded at each scheduled inspection. Sometimes frequent inspections are desirable on new equipment. The interval between inspections can then be increased at the user's discretion as he becomes more familiar with the equipment and particular operating conditions.

2. Keep on hand an adequate supply of renewal and repair parts.

3. Much time and trouble can be saved by buying renewal parts from the manufacturer who originally supplied the equipment. You are then certain such renewal parts will conform with the specifications for the original parts.

MAINTENANCE

The maintenance schedule and activity outlined in publication GET-1195 "How to Maintain Industrial Control" applies to this control.

RENEWAL PARTS

When ordering renewal parts, specify the complete catalog number of the part. If there is any doubt, include the panel catalog number and the device functional nomenclature.

TROUBLE SHOOTING

During the design and manufacture of this control, everything practical has been done to make it operate consistantly and without failure. However, eventually some trouble may develop. To be prepared, it is suggested that the user:

1. Keep the elementary diagram and instruction books handy.

2. Become familiar with the overall concept of the drive and insofar as possible with the various circuits involved.

3. Keep portable instruments handy for checking voltage, current, and resistance. A portable industrial volt - ohm - milliammeter with a 20,000 ohms per volt DC sensitivity is the minimum recommended.

4. Keep on hand an adequate supply of spare parts.

Should the drive become inoperative or its performance change, care should be exercised during trouble shooting in order not to get the drive out of adjustment. Generally, the difficulty stems from a loose connection, broken wire, or failed component and it can be corrected without making additional adjustments to the control. Considerable time will

be saved if the difficulty can be corrected without changing the component settings which necessitates retuning the drive.

The first step in the trouble shooting procedure should be to identify the problem with a specific area of the system, then systematically reduce this area to a specific module or component. The next step is to repair or replace the faulty component. The final step is to check the drive operation to be sure that no other secondary or contributing problems exist.

The following trouble shooting guide may be of assistance in tracking the problem to a general area of the control.

TROUBLE SHOOTING GUIDE

A. If drive will not start:

1. CS or ES relays may not be energized. Check 250 volt DC Exciter Bus and check protective interlocks in CS and ES coil circuits.

2. If line contactor does not pick up check run relay and contactor coil circuits.

3. If line contactor picks up but Silcomatic voltage remains at low value check Rate Reference Amplifier RRA and reference circuit on VSA modules.

B. If drive starts but goes to top speed regardless of position of handwheel controller:

1. If jog operation works check rate reference amplifier RRA.

2. Use same operation on Jog or Run and check regulator voltage (or speed) feedback circuits.

C. If drive operates ok except no current limit action check for open circuit in current limit circuit.

D. If drive coast stops from high speed when handwheel controller is turned to off check operation of speed relay CP and its auxiliary ZSR.

NOTE: *When checking the regulator module VSA or the operation of the speed relay, refer to the trouble shooting procedure in the VSA module instruction book GEK-7907 for suggested steps and normal operating values.*

APPENDIX

ISOLATED CURRENT SENSOR

SPECIFICATIONS:

Input power . . . . . 50V ± 7.5V, 50 ma max.

Output . . . . . ± 20V max., ± 8 ma if 50V power supply is used. Current rating reduces to 3 ma if 42.5V power supply is used.

Gain adjustment . . . . . Potentiometer on card provides 37 to 225 volts/volt.

Isolation . . . . . Signal input has 600 volt insulation rating to output and to power supply. Output has 250V insulation rating to power supply.

Signal input impedance . . . . . 660 ohms

Offset . . . . . ± 150 microvolts (referred to input)

Offset stability . . . . . ± 5 microvolts per °C (referred to input) and ± 200 microvolts per °C (referred to output).

Gain stability . . . . . 0.03% per °C.

Frequency response . . . . . 3 db down at 400 cycles 10° phase shift at 50 cycles

Output noise . . . . . less than 15 mv peak to peak.

APPLICATION DATA:

This package consists of current sensor amplifier 68A989866 and isolation amplifier 68A989827 plus a filter network.

The signal input to the isolated current sensor will normally be taken from a 100 millivolt shunt. The amplifier gain is adjustable and should normally be adjusted to give 15 volts output at the maximum expected input signal. Each 1/10 revolution of the gain pot will give a change of approximately 36 millivolts in the amount of input signal required for 15 volts output.

The isolated current sensor prototypes provided more than 90 db rejection of 60 cycle common mode voltage. Since the amplifier has a high gain the input wires must be protected against noise pick-up. Twisted input leads will give satisfactory protection if the input leads are 6 feet or less in length. Output lines should be shielded twisted and should be run in a level 1 wire tray unless the isolator is page mounted.

When the input power is taken from a 250 volt exciter bus, it is recommended that a 2000 ohm resistor be connected in series with each input line. This will help reduce common mode input from the power source, and will also limit the fault current if one of the input power terminals should accidentally be grounded.

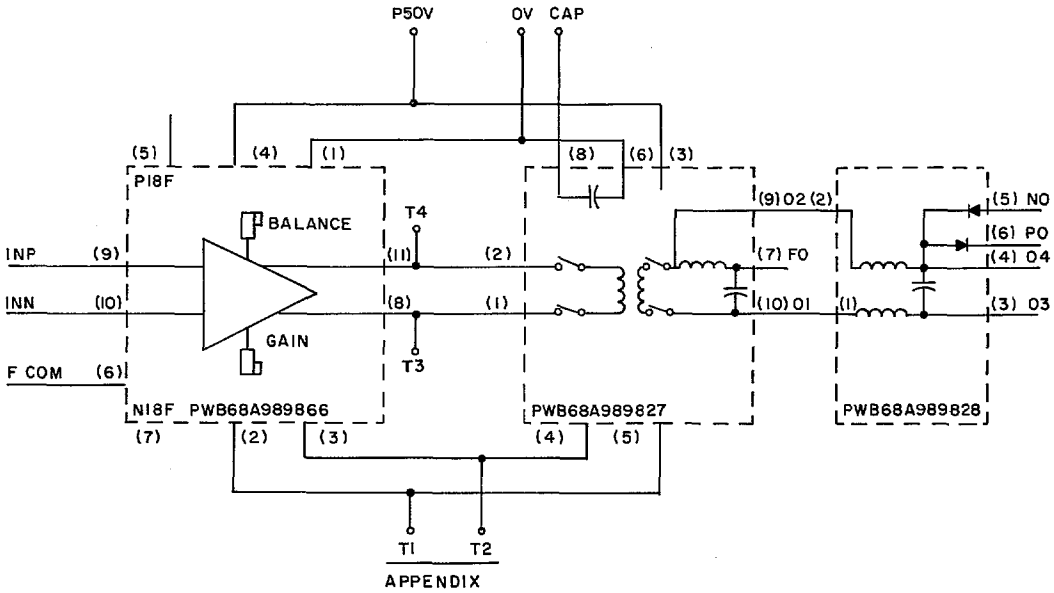


Fig. 6. Isolated current sensor

INDUSTRY CONTROL DEPARTMENT

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